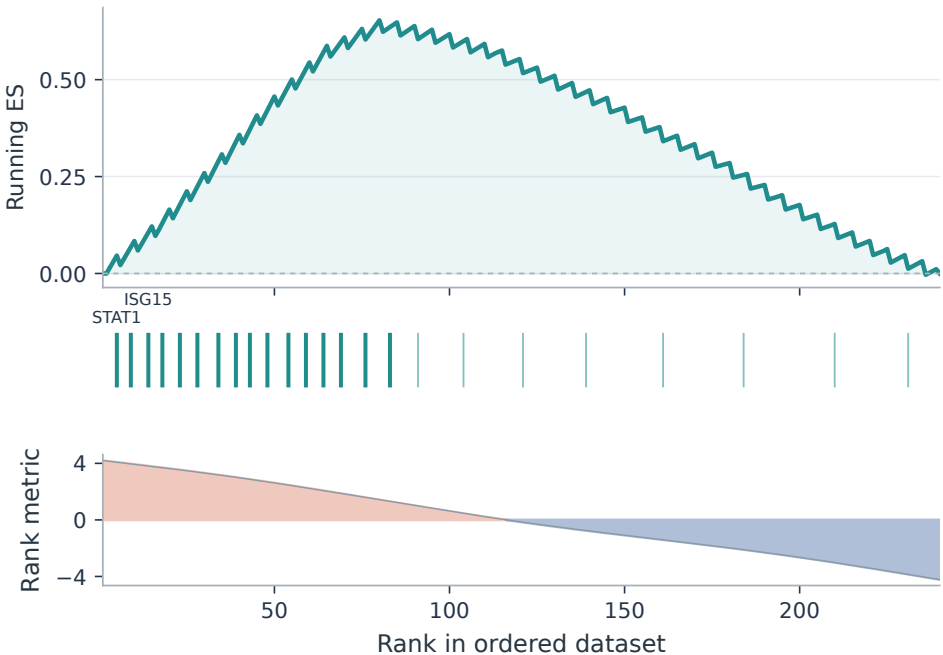
Gene set enrichment

Illustrative condition B versus A / complete ranked universe: 240



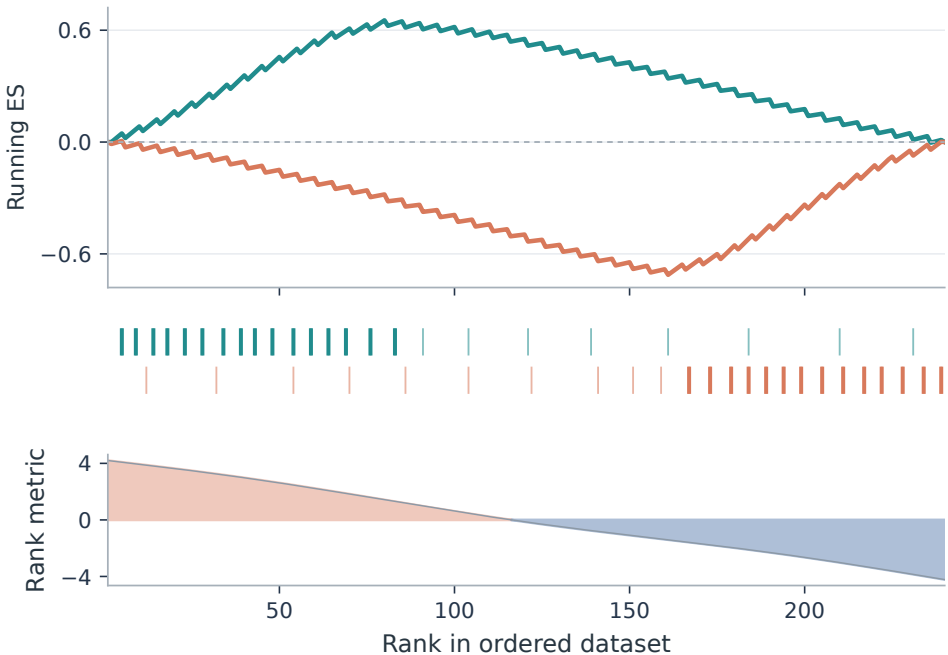
Interferon response
NES +1.92
FDR 0.0008

Gene-set hits
Bold: supplied leading edge

Synthetic gene sets v2 (not MSigDB); ES, NES, adjusted probabilities and hit positions supplied unchanged.

Enrichment programmes · overlay

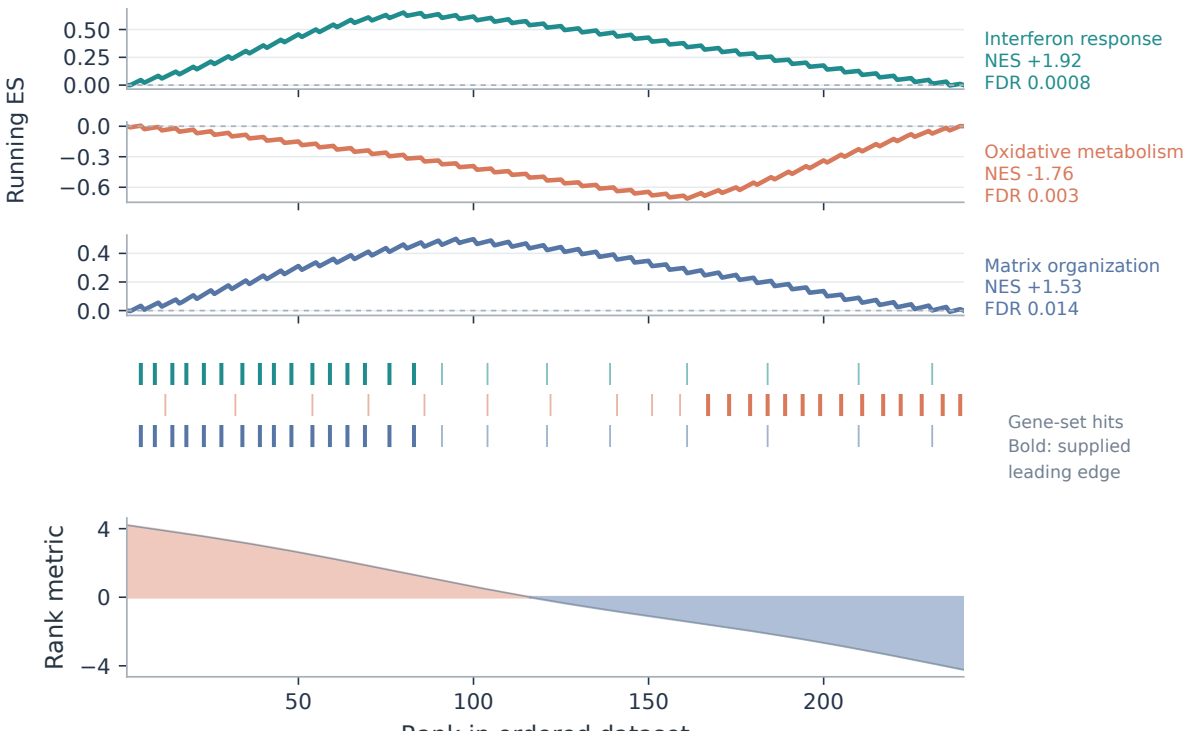
Illustrative condition B versus A / complete ranked universe: 240



Synthetic gene sets v2 (not MSigDB); ES, NES, adjusted probabilities and hit positions supplied unchanged.

Enrichment programmes · facets

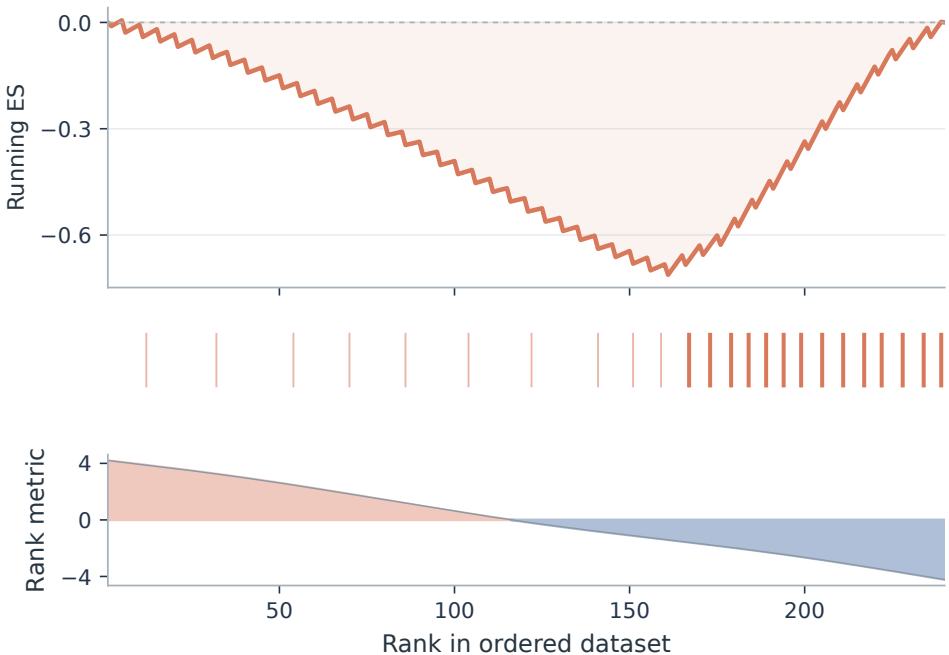
Illustrative condition B versus A / complete ranked universe: 240



Synthetic gene sets v2 (not MSigDB); ES, NES, adjusted probabilities and hit positions supplied unchanged.

Gene set enrichment

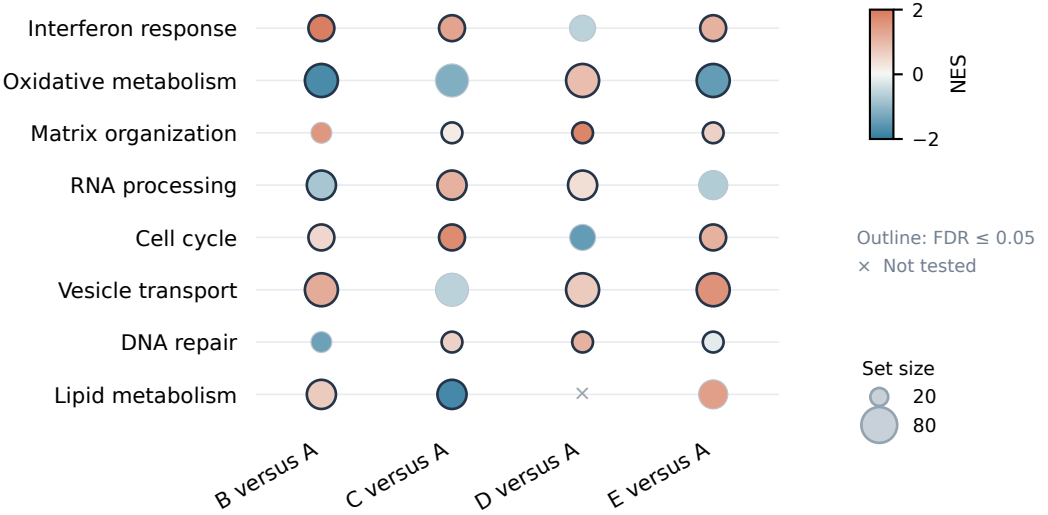
Illustrative condition B versus A / complete ranked universe: 240



Synthetic gene sets v2 (not MSigDB); ES, NES, adjusted probabilities and hit positions supplied unchanged.

Enrichment across contrasts

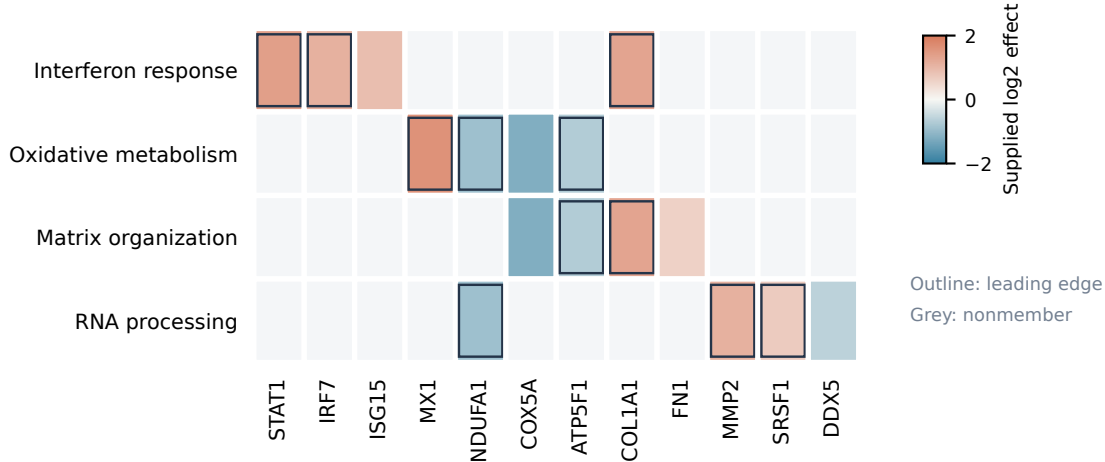
Colour: normalized enrichment / Area: gene-set size



Selected terms and contrasts retained; FDR and NES are separate supplied channels.

Leading-edge membership

Outlined cells identify the supplied leading-edge subset



Membership and gene effects supplied; no gene selection, clustering or inference.

Over-represented programmes

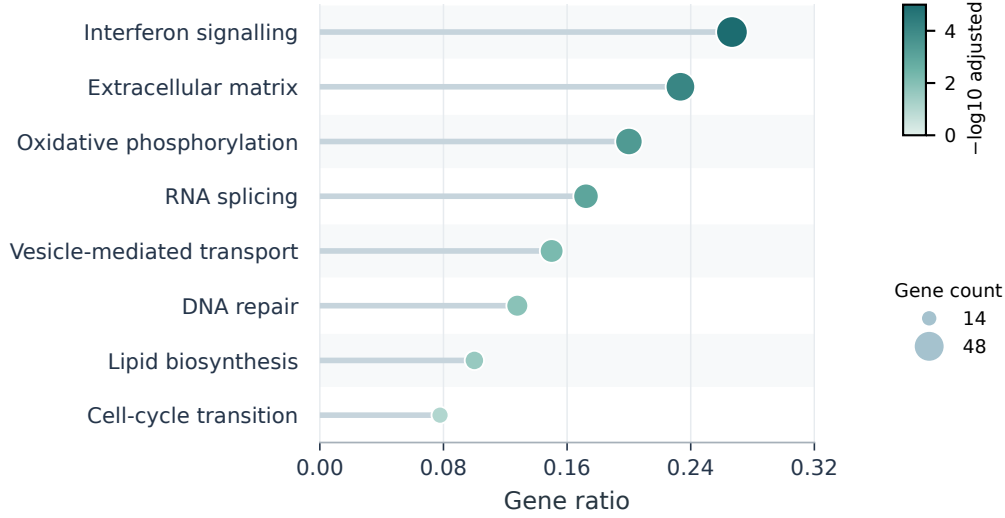
Selected query n=180 / Synthetic universe n=1,000



Ratio = count / selected query; background and adjusted probabilities are explicit frozen inputs.

Over-represented programmes

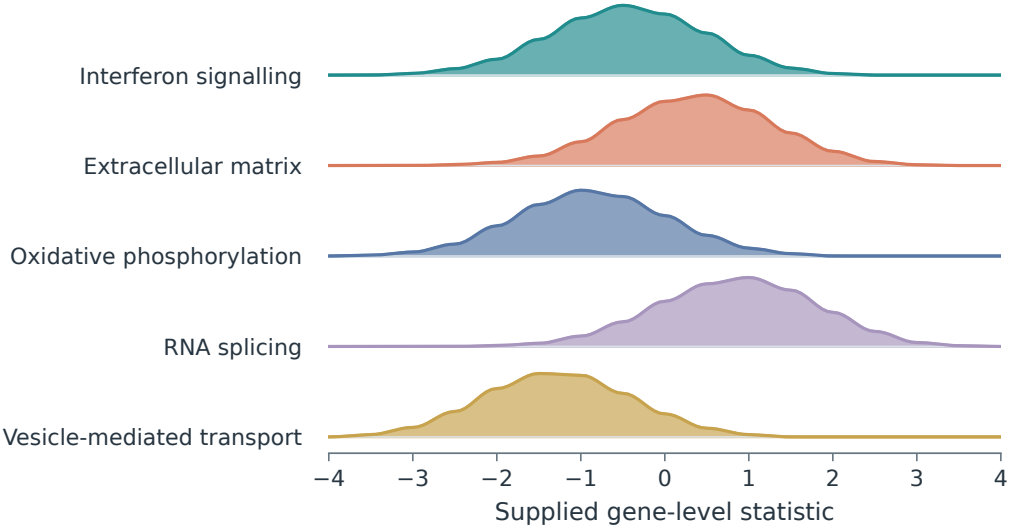
Selected query n=180 / Synthetic universe n=1,000



Ratio = count / selected query; background and adjusted probabilities are explicit frozen inputs.

Core enrichment distributions

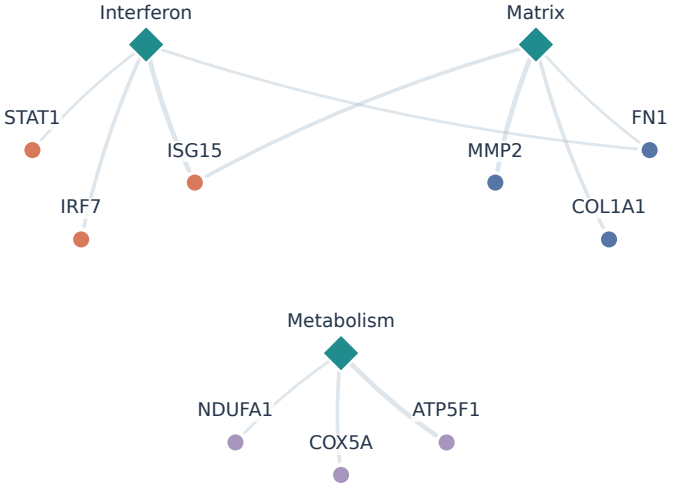
Illustrative frozen core-gene density profiles



Common height scale; supplied density profiles only. No KDE or core-gene selection performed.

Genes connecting biological programmes

membership / Explicit gene-to-programme membership



Programmes
Response
Matrix
Energy

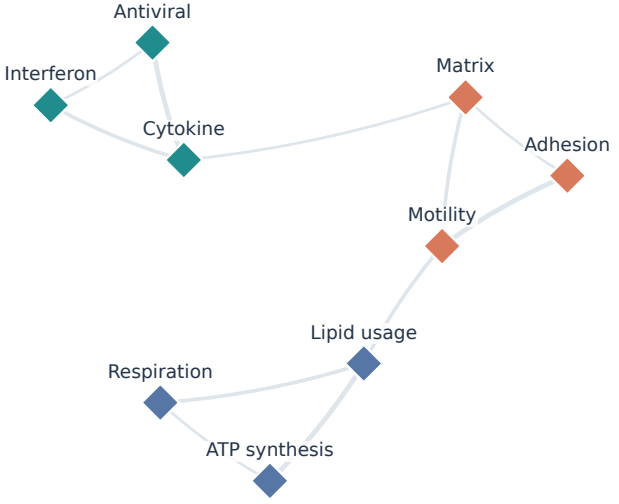
Diamonds: terms
Circles: genes

Edge weight
0.35
0.8

Node area
Fixed display area

Enrichment map · selected programmes

overlap / Supplied gene-set similarity



Immune
Matrix
Energy

Diamonds: terms

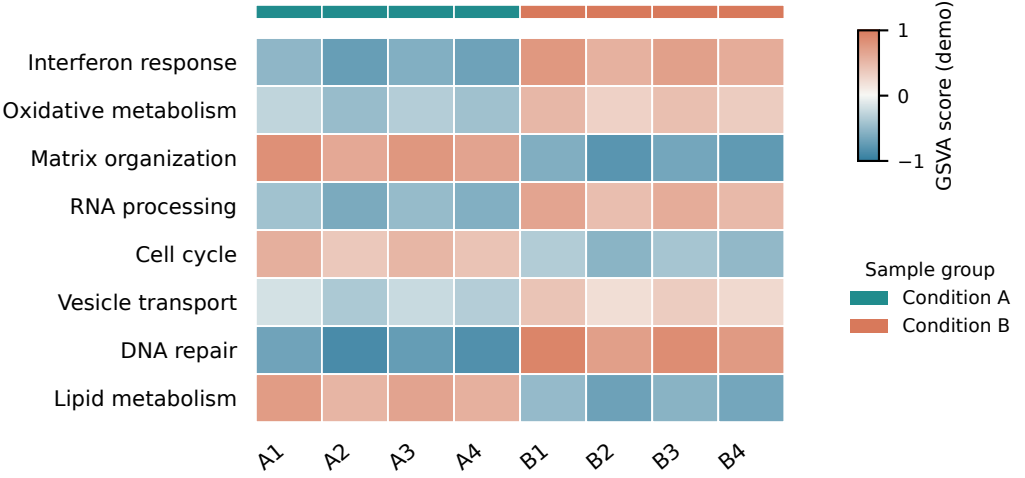
Edge weight
0.35
0.8

Node area
Fixed display area

Topology, coordinates and weights supplied. No layout optimization, edge inference or causal validation.

Pathway scores across samples

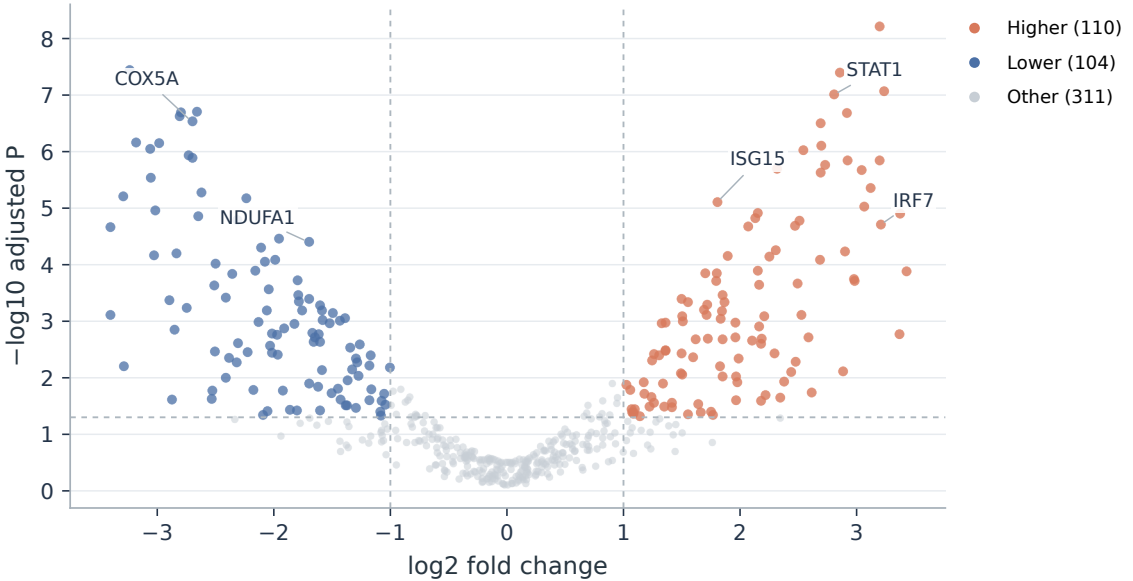
Illustrative frozen GSVA score matrix / no z-scaling



Values and identities retained. No scaling, sample clustering or upstream activity estimation.

Transcriptome response

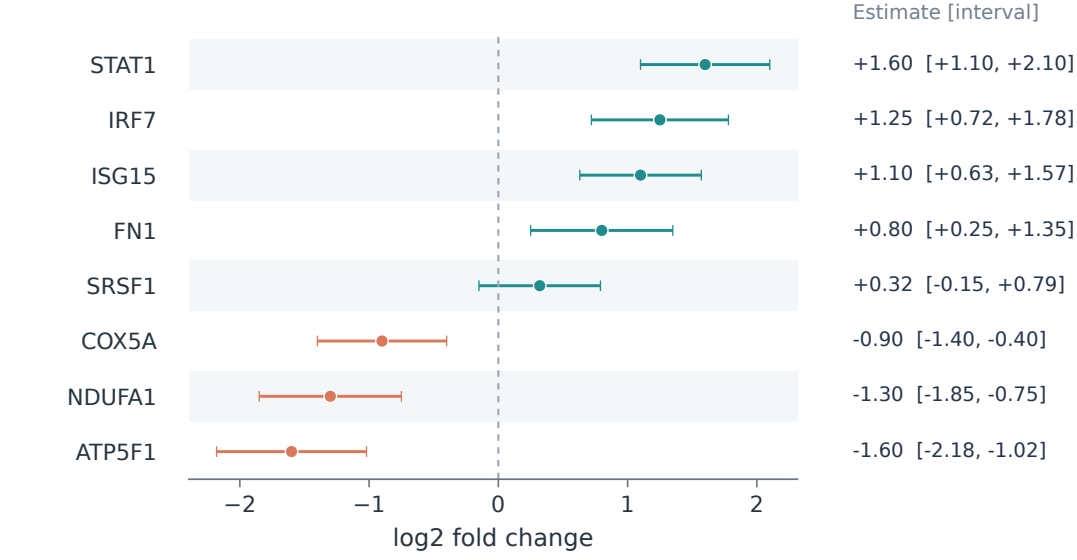
Illustrative DESeq2 result export / condition B versus A



Visual thresholds only; exact supplied fold changes and adjusted probabilities. Label selection declared.

Gene-level differential effects

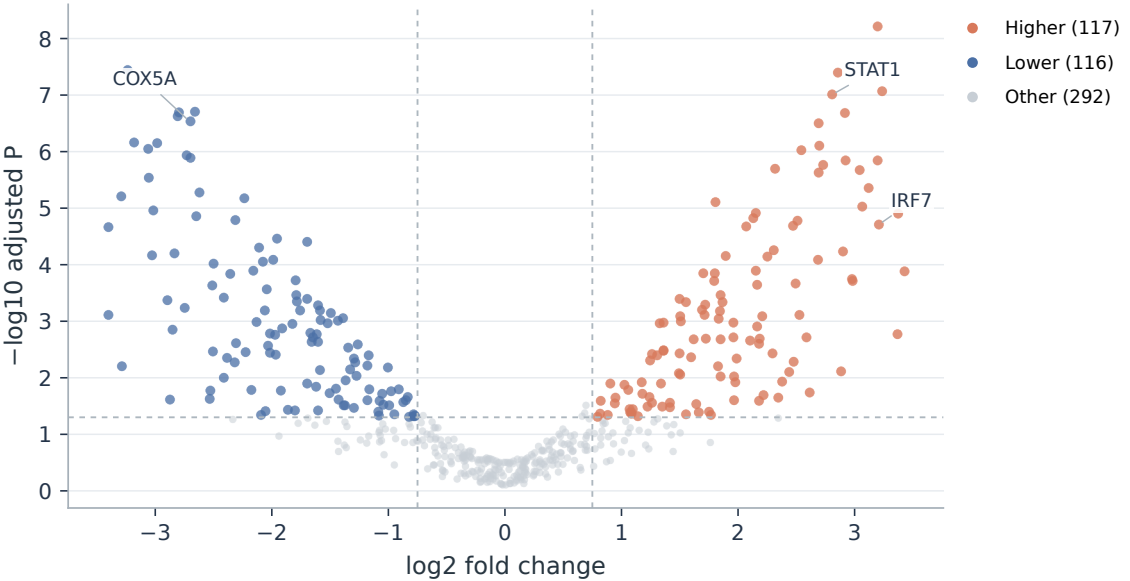
Illustrative limma export / supplied 95% confidence intervals



Existing intervals only; no model fit, empirical-Bayes moderation or uncertainty estimation.

Protein abundance response

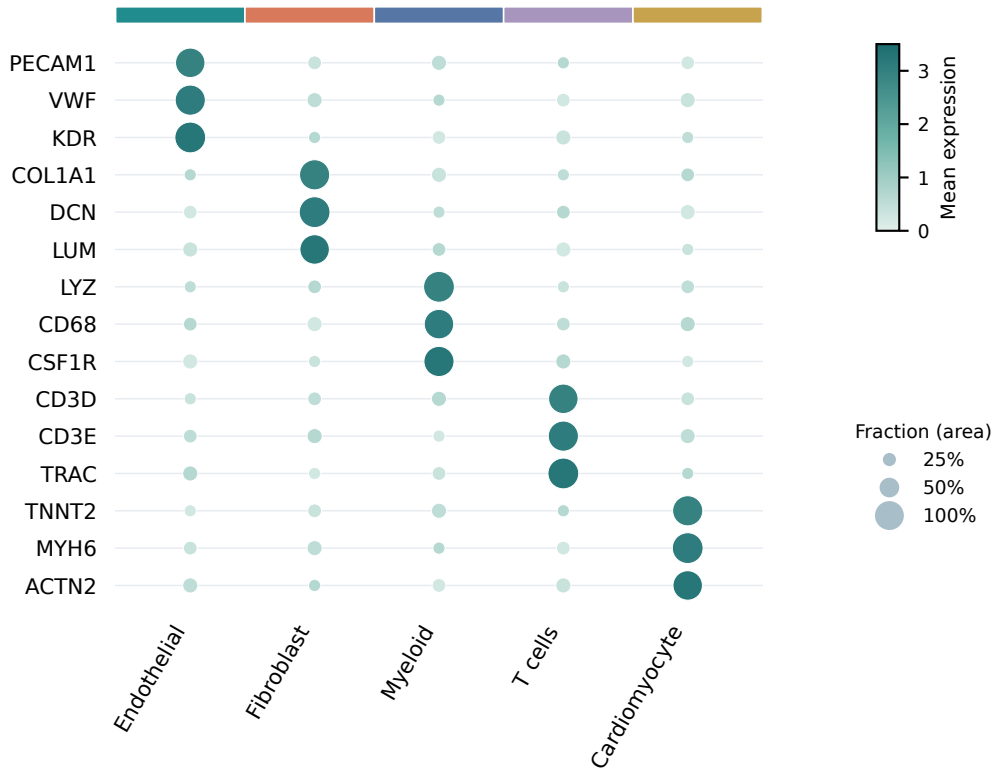
Illustrative MSstats result export / fixed protein-level effects



Visual thresholds only; exact supplied fold changes and adjusted probabilities. Label selection declared.

Marker expression by cell identity

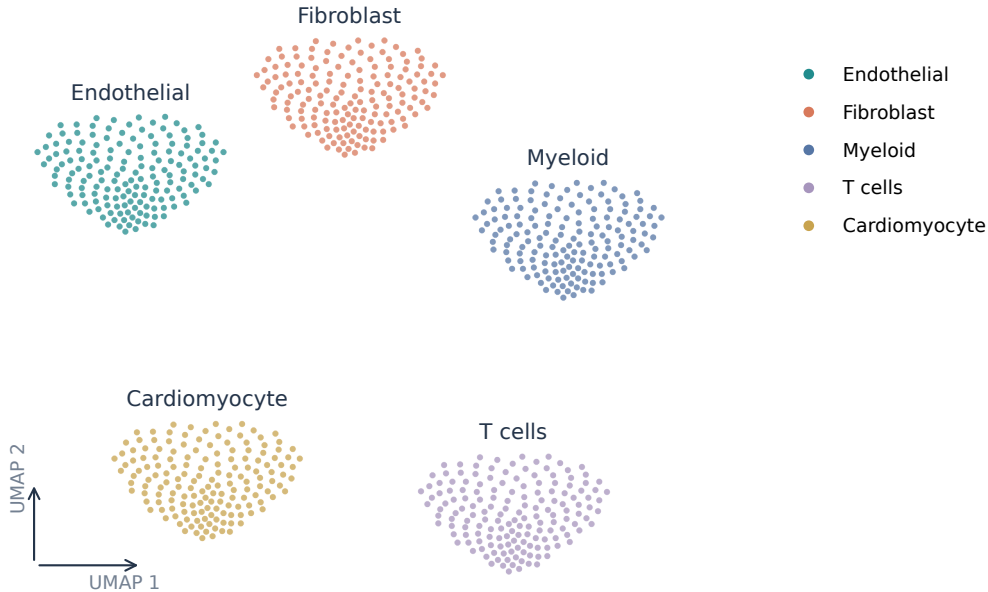
Fraction denominator: all displayed cells in each identity (synthetic)



Means and fractions supplied. No expression normalization, clustering or independent-subject inference.

Cell identities · Seurat

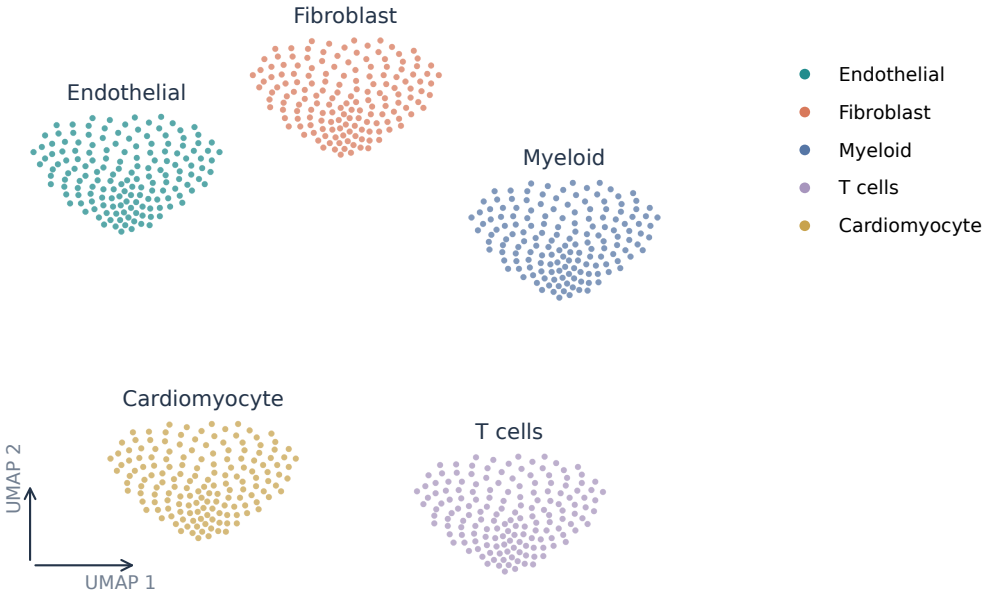
Illustrative frozen embedding geometry / not a fitted UMAP



Stored coordinates and explicit label anchors; no embedding fit, density estimate or donor-level inference.

Cell identities · Scanpy

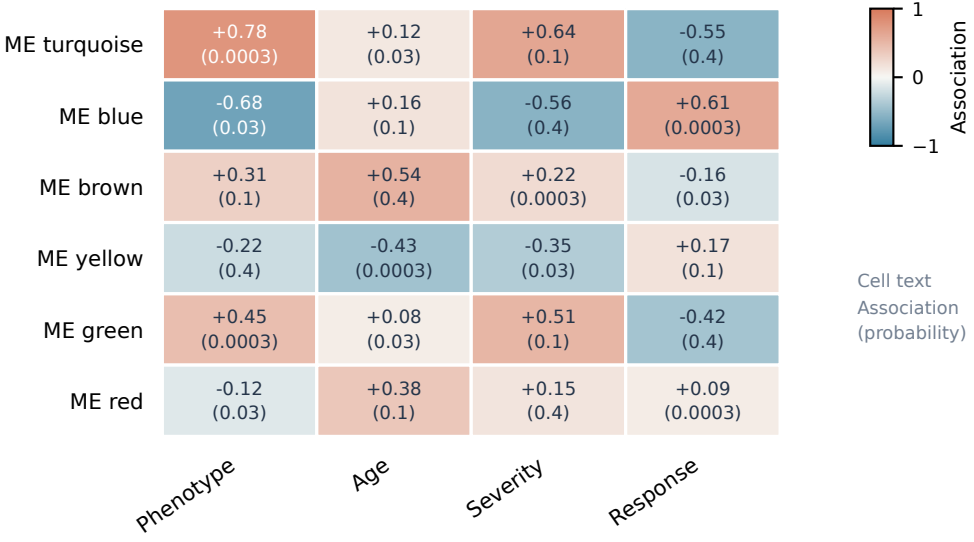
Illustrative frozen embedding geometry / not a fitted UMAP



Stored coordinates and explicit label anchors; no embedding fit, density estimate or donor-level inference.

Module–trait relationships

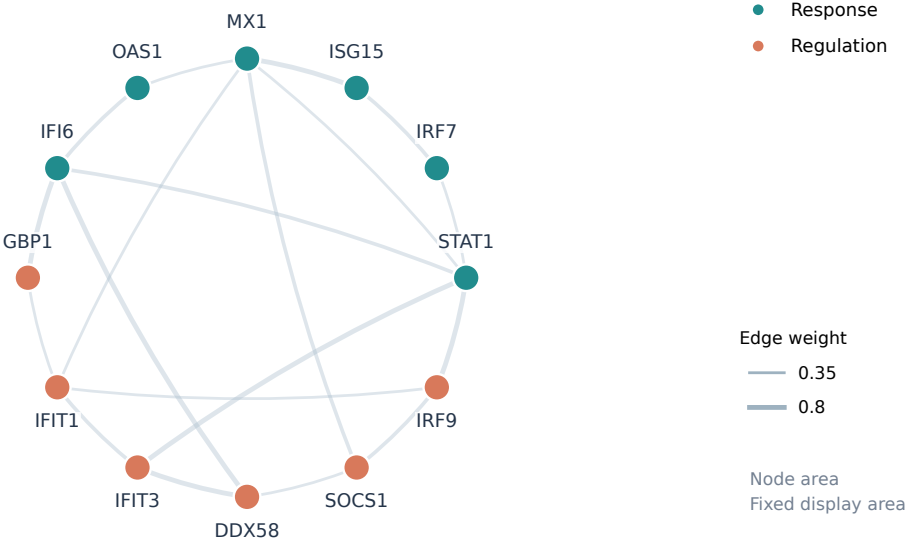
Illustrative donor-level module–trait correlation matrix



Parentheses: supplied unadjusted probabilities (demo); module identities and independent sample context retained.

Co-expression module · response genes

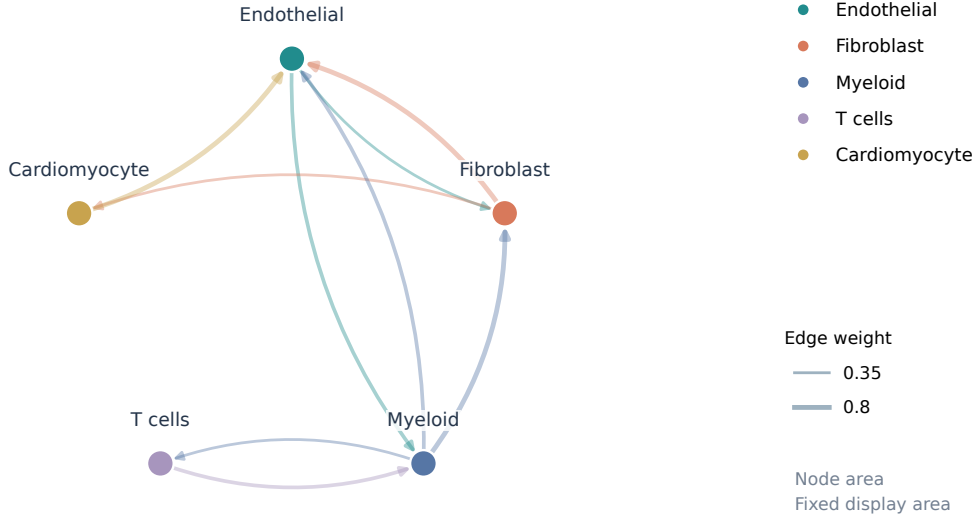
correlation / Selected weighted co-expression edges



Topology, coordinates and weights supplied. No layout optimization, edge inference or causal validation.

Predicted cell–cell communication

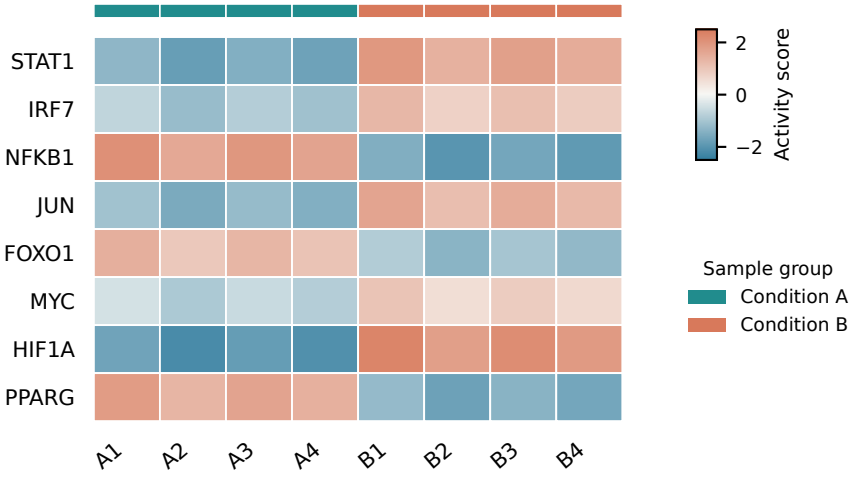
predicted communication / Supplied aggregate communication weight



Topology, coordinates and weights supplied. No layout optimization, edge inference or causal validation.

Transcription factor activity

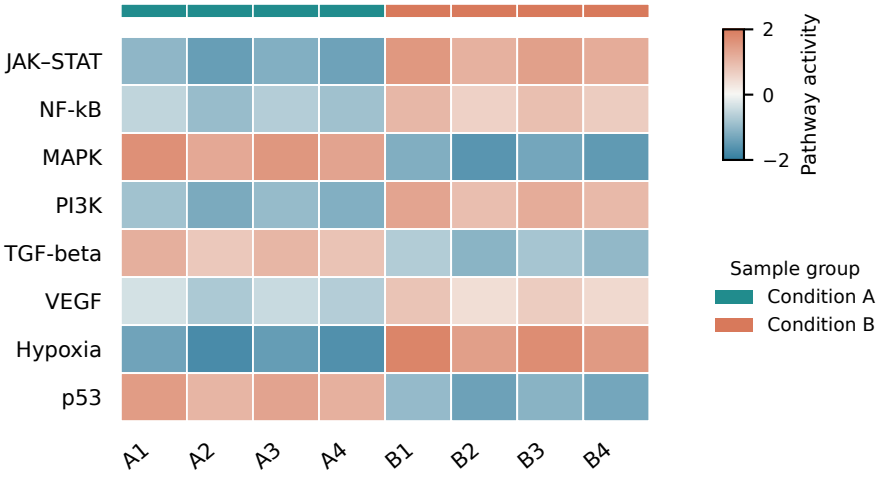
Illustrative decoupleR export / supplied signed activity scores



Values and identities retained. No scaling, sample clustering or upstream activity estimation.

Signalling pathway activity

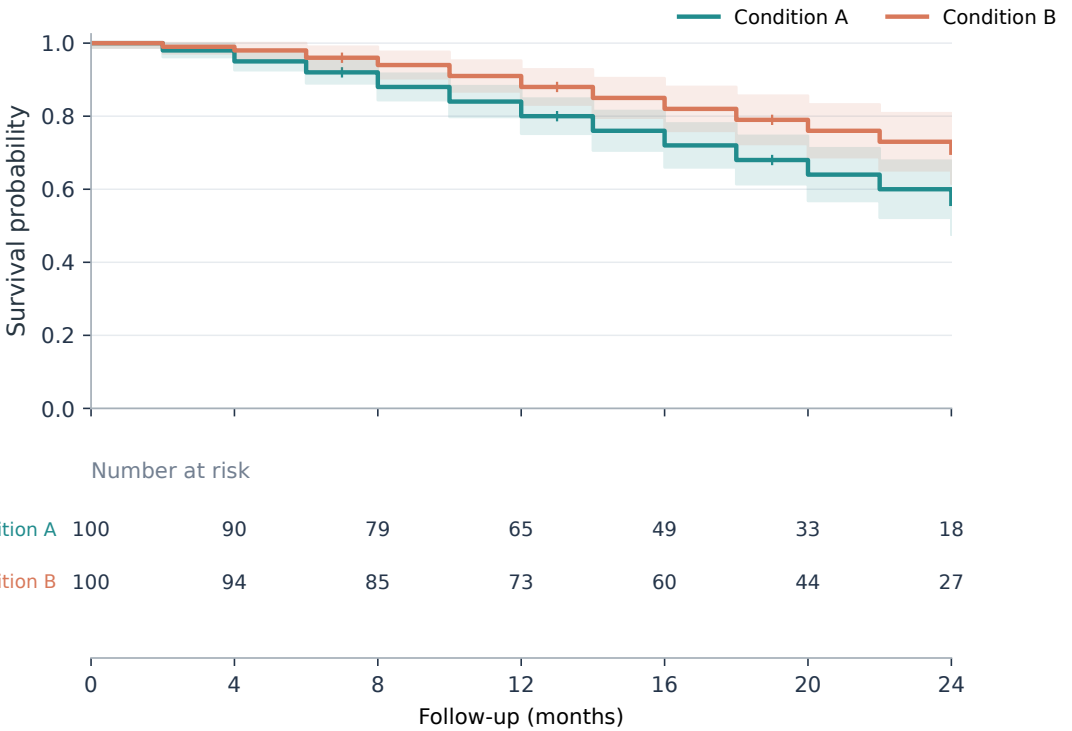
Illustrative PROGENy export / fixed score units and sample order



Values and identities retained. No scaling, sample clustering or upstream activity estimation.

Survival over follow-up

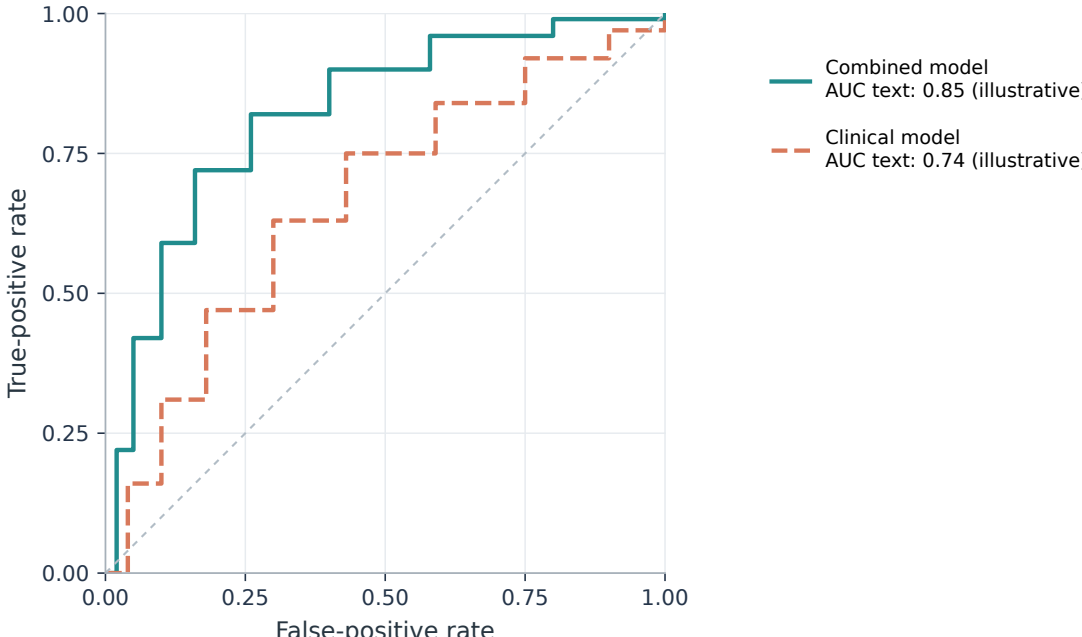
Synthetic outcome curves / no patient data or survival fitting



Illustrative supplied 95% confidence bands; risk convention: start_of_period. No survival fit or log-rank test.

Model discrimination

Illustrative held-out ROC coordinates / no model training



Curves and optional AUC text supplied by the analysis host; no AUC or uncertainty calculation.